1. Movie Ticket Booking System

Scenario

A movie theater wants to calculate the ticket price based on different booking options. Since customers may provide different details while booking, use **method overloading**.

Requirements

Create a class MovieTicket and overload a method named bookTicket().

Implement the following methods:

1. bookTicket(int tickets)
    - Calculate the total amount for the given number of tickets.
    - Assume the price of one ticket is ₹200.
2. bookTicket(int tickets, boolean isPremium)
    - If isPremium is true, ticket price is ₹350 each.
    - Otherwise, ticket price is ₹200 each.
3. bookTicket(int tickets, boolean isPremium, double discount)
    - Calculate the total amount.
    - Apply the given discount percentage.
    - Display the final payable amount.

Write a main() method to call all three overloaded methods.

2. Electricity Bill Calculator

Scenario

An electricity department calculates bills based on different customer inputs.

Requirements

Create a class ElectricityBill and overload a method named calculateBill().

Implement the following methods:

1. calculateBill(int units)
    - Charge ₹8 per unit.
    - Display the total bill.
2. calculateBill(int units, double serviceCharge)
    - Calculate the bill amount.

    - Add the service charge.
    - Display the final bill.
3. calculateBill(int units, double serviceCharge, double tax)
    - Calculate the bill amount.
    - Add the service charge.
    - Apply the given tax percentage.
    - Display the final amount to be paid.

Write a main() method to call all three overloaded methods with different values.

Constructor Overloading Scenarios (Practice Questions)

1. Online Shopping Product

An online shopping application stores product details. Sometimes only the product name is available, sometimes the product name and price are available, and sometimes the product name, price, and category are available. Design a Product class using **constructor overloading** to initialize the object in these different situations. Create objects using all constructors and display the product details.

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2. Student Admission System

A college admission system stores student information. During the admission process, some students provide only their name, while others provide their name and course, and some provide their name, course, and fee details. Design a Student class using **constructor overloading** to initialize student information based on the available data. Display the details of each student.

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3. Bank Account Creation

A banking application allows customers to create accounts in different ways. Some customers create an account with only their name, some provide their name and account type, and others provide their name, account type, and initial deposit. Design a BankAccount class using **constructor overloading** to initialize customer details accordingly. Display the account information.

Method Overloading Scenarios (Practice Questions)

1. Electricity Bill Calculator

An electricity billing system calculates the bill in different ways. If only the number of units is provided, calculate the bill using the default rate. If the units and rate per unit are provided, calculate using the given rate. If units, rate, and tax percentage are provided, calculate the final bill including tax. Design a class using **method overloading** to perform these calculations.

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2. Employee Salary Calculator

An HR management system calculates employee salaries in different situations. Calculate the salary using only the basic salary, using the basic salary and bonus, and using the basic salary, bonus, and overtime amount. Implement this using **method overloading** and display the calculated salary.

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3. Courier Charge Calculator

A courier company calculates delivery charges differently based on the available information. Calculate the delivery charge using only the package weight, using the package weight and delivery distance, and using the package weight, delivery distance, and express delivery option. Design a class using **method overloading** to calculate the delivery charge for each case.

combination of constructor overloading and method overloading

1. Hotel Room Booking System

A hotel booking application manages room reservations for its customers.

Design a `RoomBooking` class using both **constructor overloading** and **method overloading**.

Requirements

- Create constructors to initialize booking details in the following ways:
  - Customer name only.
  - Customer name and room type.
  - Customer name, room type, and number of days.
- Create overloaded `calculateBill()` methods to:
  - Calculate the bill using the default room rate.
  - Calculate the bill using a custom room rate.
  - Calculate the bill using a custom room rate and a discount percentage.
- Create a `displayBooking()` method to display all booking details.

2. Mobile Recharge System

A mobile recharge application allows users to recharge their mobile numbers.

Design a `Recharge` class using both **constructor overloading** and **method overloading**.

Requirements

- Create constructors to initialize recharge details in the following ways:
  - Mobile number only.
  - Mobile number and recharge amount.
  - Mobile number, recharge amount, and service provider.
- Create overloaded `recharge()` methods to:
  - Recharge with the default validity.
  - Recharge with a custom validity.
  - Recharge with a custom validity and a promotional cashback amount.
- Create a `displayRechargeDetails()` method to display all recharge information.